Properties — supporting figures

kaizen step 1.3.0 — MarkdownToLaTeX 1.0.0

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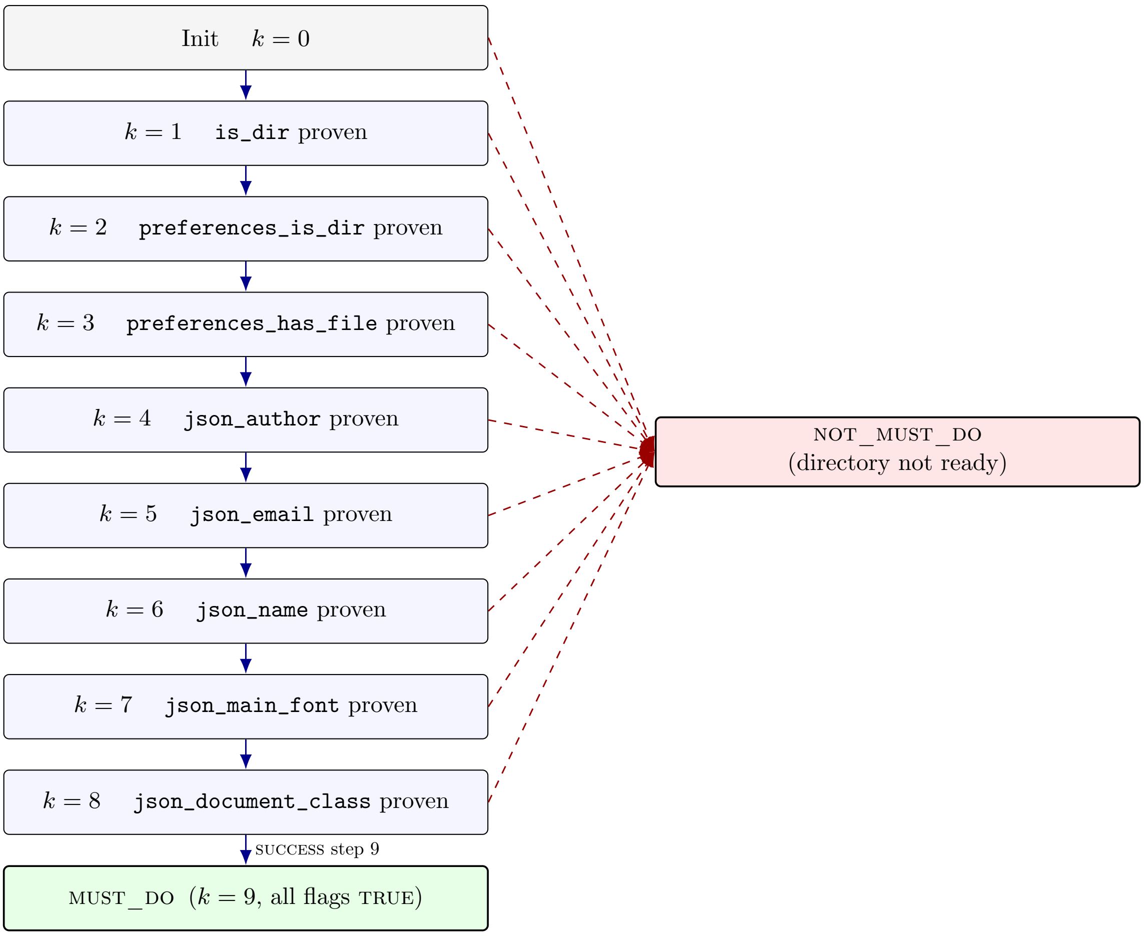
Figure 1 — State-transition diagram of `Directory.tla`

The procedure threads a fixed chain of nine checks. From every intermediate state, exactly two non-stutter transitions are enabled: the (success) step that proves the next flag and advances along the chain, and the (failure) step that latches `verdict := NOT_MUST_DO`. The terminal state `MUST_DO` is only reachable through the ninth and final success.

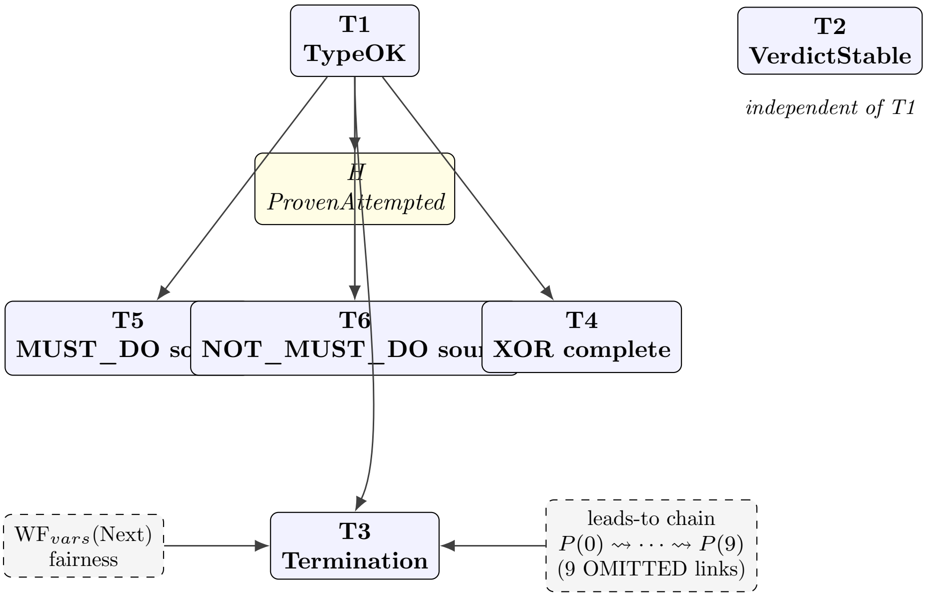


Solid blue arrows: SUCCESS *actions (advance one rung along the chain). Dashed red arrows:* FAIL(s) *actions (latch* NOT_MUST_DO *from any rung). The two terminal states are absorbing: only stuttering is enabled once entered.*

Reading note. Each state S_k has `verdict = "UNKNOWN"` and exactly k proof flags set. The two terminal states MUST_DO and NOT_MUST_DO are absorbing: once entered, only stuttering is enabled (theorem T2, verdict stability, in `Directory.1.3.0.tla.txt`).

Figure 2 — Proof-dependency DAG over T1–T6

T1 (type-correctness) underpins every other proof in the module. T2 (verdict stability) stands on its own — its proof depends only on the action structure, not on TypeOK. The helper lemma H (`ProvenAttemptedInvariant`) bridges proof flags and `attempted`, and is consumed only by T6. T3 (termination) also draws on the fairness clause $\mathrm{WF}_{vars}(\mathit{Next})$ and on the leads-to chain of nine OMITTED links discharged interactively in the TLA Toolbox.



Reading note. Solid arrows are proof dependencies: the head theorem is invoked in the body of the tail theorem's TLAPS proof. Yellow node H is a helper lemma stated and proved in the same module; dashed boxes on either side of T3 are external assumptions (fairness from `Spec`, and the WF1 leads-to chain whose nine links are flagged `OMITTED` in `Directory.1.3.0.tla.txt`, to be discharged in the TLA Toolbox).

Figure 3 — Hasse-style chain of the `Predecessors` ordering

The relation `Predecessors`(s) is total: every step has every earlier step as a strict predecessor and none as an incomparable peer. The Hasse diagram is therefore a single linear chain. Step 9 (`json_document_class_legal`) is the unique greatest element and is also the only step whose success transitions `verdict` to `MUST_DO`.

Reading note. An arrow $a \to b$ means $a \in \mathit{Predecessors}(b)$. Read upward, this is also the temporal order in which proof flags can flip from FALSE to TRUE: a flag is provable only after every flag below it has already been proved. This monotone chain is what makes the inductive arguments behind T5 and T6 fall out cleanly.